

Week 6 Workshop

Question 1

ProShares claims that SDS (ticker symbol: SDS) aims to provide approximately -2 times the daily return of the S&P 500 Index (^GSPC). Using the concepts learned in this unit, explain how you would evaluate whether ProShares UltraShort S&P500 (SDS) appropriately tracks its stated daily target.

To do this, download daily adjusted closing price data for SDS and daily closing price data for the S&P 500 Index (^GSPC) from Yahoo Finance, using a sample period that starts on 13 July 2006.

Question 2

Using Vanguard style-box ETFs to illustrate size and value effects. Consider the following US-listed Vanguard ETFs arranged in a 3×3 style box:

	Growth	Blend	Value
Small	VBK	VB	VBR
Mid	VOT	VO	VOE
Large	VUG	VV	VTV

Use monthly adjusted closing prices downloaded from Yahoo Finance for the period December 2015 to December 2025. Compute log returns, in percentage terms per month, for all nine ETFs. Then construct a 3×3 table showing the sample mean monthly log return (%) for each ETF in the style-box. Briefly explain what each dimension of the table represents.